Adı Soyadı:
Numarası:

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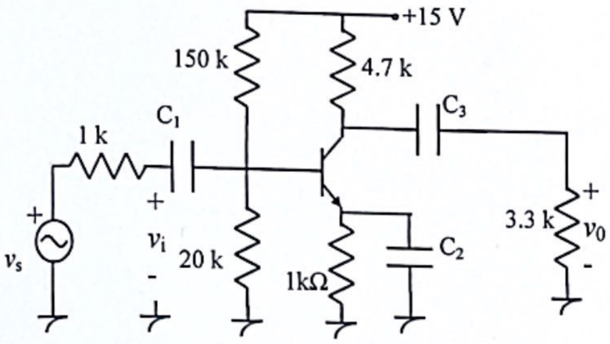
1	2	3	4

Toplam:

Tarih:24.04.2022 **Süre:**90 dak.BAŞARILAR… Doç.Dr. Murat CEYLAN & Prof.Dr.Seral ÖZŞEN

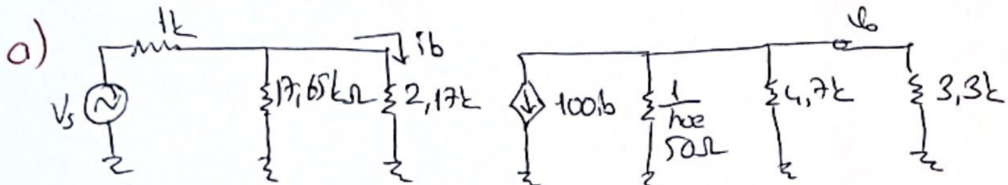
S.1. (25 Puan)



h_{fe}=100, h_{ie}=2,17k Ω, hoe = 20m S olduğuna göre;

a) Devrenin AC eşdeğerini çiziniz.
b) Z_i=?, Z_o=?
c) A_V=?, A_{vs}=? A_i=?
d) v_s=1mV tepe değerine sahip sinüzoidal bir sinyal ise v_s ve v_0 sinyallerinin dalga şeklini çiziniz.

a)



(3puan)

b) $Zp = 17,65k // 2,17k$

$\boxed{Zp = 1,93k\Omega}$ (2p)

$Z_o = 4,7k // 50\Omega$

$\boxed{Z_o = 49,47\Omega}$ (2p)

Not: $\frac{1}{hoe}$ = 50kΩ alalırsanız da $Z_o = 4,29k\Omega$ (puan kırılmayacak)

(5p)

c) $A_V = \frac{V_o}{V_i} = \frac{-100 i_b . (50\Omega // 4,7k // 3,3k)}{2,17k . i_b} \Rightarrow \boxed{A_V = -2,24}$

$\boxed{A_V = -85,95}$

(5p)

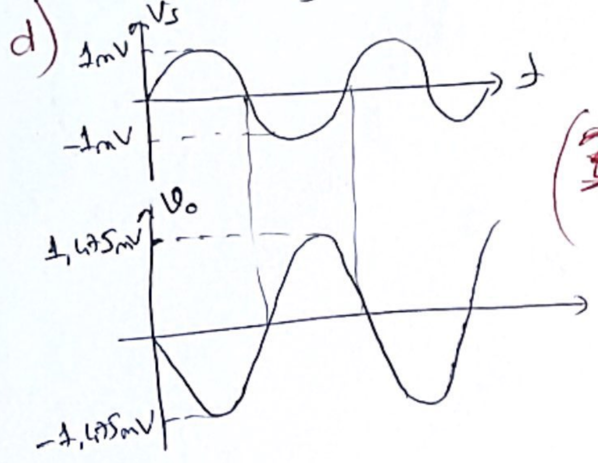
$A_{VS} = A_V . \frac{Z_i}{Z_i + Z_s} = -2,24 . \frac{1,93k}{2,93k} \Rightarrow \boxed{A_{VS} = -1,475}$

$\boxed{A_{VS} = -56,61}$

$A_i = -A_V . \frac{Z_i}{R_y} = \boxed{1,31}$ (5p)

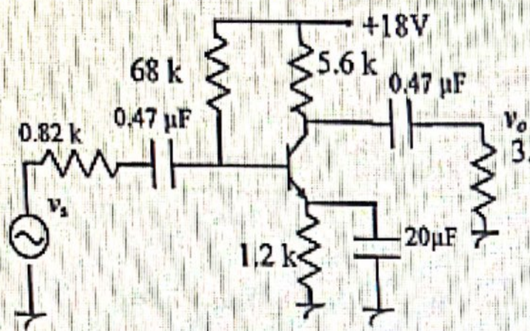
$\boxed{A_i = 50,27}$

d)



(3p)

S.2 (25 Puan)



$\beta = h_{fe} = 120$, $r_e = 28.48\Omega$, $C_{bc} = 12pF$, $C_{be} = 40pF$, $C_{ce} = 8pF$, $C_{w1} = 5pF$, $C_{w2} = 8pF$ ise;

a) $A_v = ?$

b) $f_{alt} = ?$

c) $f_{üst} = ?$

a) $A_v = \dfrac{-h_{fe}.(3.3k \parallel 5.6k)}{h_{ie}}$

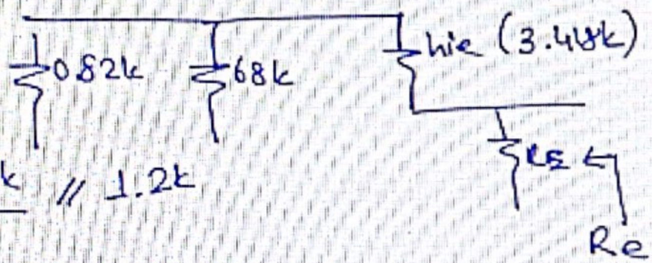
$\boxed{A_v = -72.91}$ 5 puan

b) $Z_i = 68k \parallel \underbrace{120.(28.48)}_{3.418k} \Rightarrow Z_i \cong 3.25k$

$f_S = \dfrac{1}{2\pi(R_S + Z_i).C_S} = \dfrac{1}{2\pi(0.82k + 3.25k)(0.47\mu)} = 83.24\,Hz$ 3

$f_C = \dfrac{1}{2\pi(Z_o + R_L).C_C} = \dfrac{1}{2\pi(5.6k + 3.3k)(0.47\mu)} = 38.05\,Hz$ 3

$f_E = \dfrac{1}{2\pi R_e C_E}$



$R_e = \dfrac{(0.82k \parallel 68k) + 3.418k}{120} \parallel 1.2k$

$R_e = \dfrac{0.81k + 3.418k}{120} \parallel 1.2k = 34{,}22\,\Omega$ 1

$f_E = \dfrac{1}{2\pi.(34.22)(20\mu)} = 232.66\,Hz$ 3

$\boxed{f_a = 232{,}66\,Hz}$

c) $C_i = C_{w1} + C_{be} + C_{bc}(1 - A_v) = 5pF + 40p + 12p(1 - (-72.91)) = 931.92p$

$R_i = 0.82k \parallel 68k \parallel 3.418k = 654.79\,\Omega$

$f_i = \dfrac{1}{2\pi C_i.R_i} = \dfrac{1}{2\pi.(931.92)10^{-12}.(654.79)} = 260{,}95\,kHz$ 4

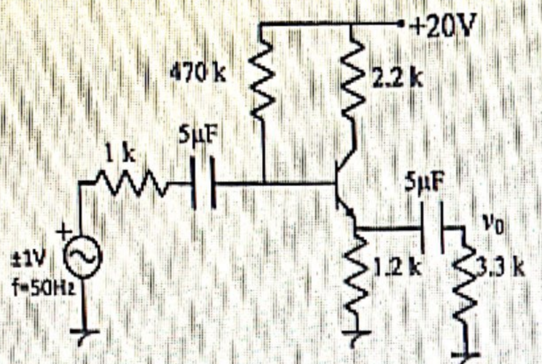
$C_o = C_{w2} + C_{ce} + C_{bc}(1 - \frac{1}{A_v}) = 8 + 8 + 12 \cong 28p$

$R_o = 3.3k \parallel 5.6k = 2.08k$

$f_o = \dfrac{1}{2\pi C_o.R_o} = \dfrac{1}{2\pi.(28.10^{-12})(2.08.10^3)} = 2.73\,MHz$ 4

2 $\boxed{f_ü = 260.95\,kHz}$

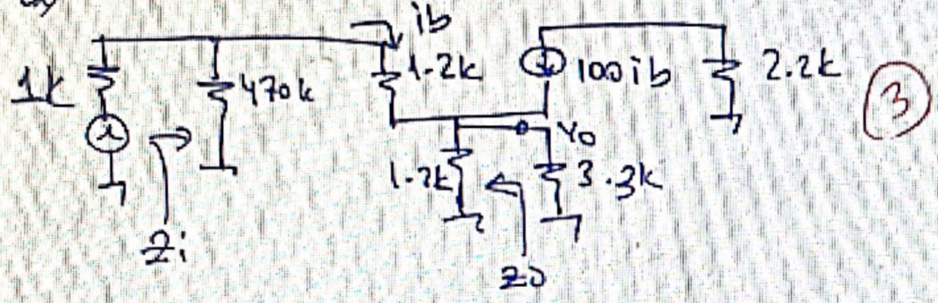
S.3 (25 puan)



h_{ie}=1.2kΩ, h_{fe}=β=100 olarak verilmektedir. Buna göre;

a) AC eşdeğer devreyi çiziniz.
b) Z_i=?
c) Z_o=?
d) A_v=?
e) A_i=?
f) A_{vs}=?
g) v_o sinyalinin dalga şeklini çiziniz.

a)



3

b) $Z_B = 1.2k + 101(1.2k // 3.3k) = 90.08k$

$Z_i = 470k // 90.08k \cong 75.6k$ $\boxed{Z_i \cong 75.6k}$ 4

c) $Z_o = \dfrac{(1k // 470k + 1.2k)}{101} // 1.2k$

$Z_o = \dfrac{2.2k}{101} // 1.2k \Rightarrow \boxed{Z_o \cong 21.4\Omega}$ 4

d) $V_o = 101.ib\,(1.2k // 3.3k) = 0.88k.\,101.ib = 88.88k\,ib$

$V_i = ib.\,Z_B = 90.08k.ib$ 4

$A_v = \dfrac{V_o}{V_i} \Rightarrow \boxed{A_v \cong 0.987}$ 4

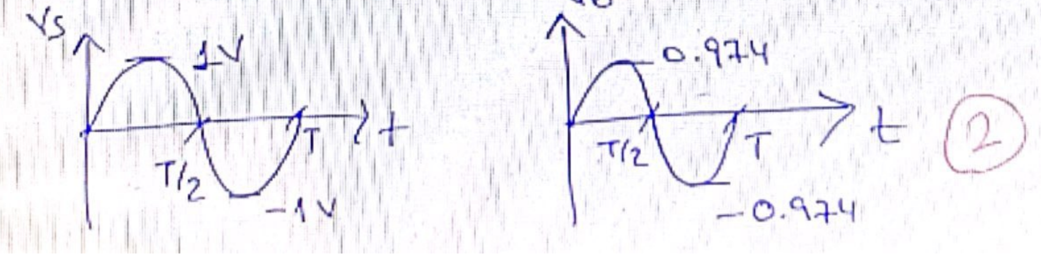
f) $A_{vs} = A_v.\dfrac{Z_i}{Z_i + R_s} = 0.987.\dfrac{75.6k}{76.6k} \Rightarrow \boxed{A_{vs} = 0.974}$ 4

e) $I_o = -\dfrac{V_o}{3.3k} = \dfrac{-88.88k\,ib}{3.3k} = -26.93k\,ib$

$I_i = ib + \dfrac{V_i}{470k} = ib + \dfrac{(90.08k)ib}{470k} = 1.19ib$

$\boxed{A_i = \dfrac{I_o}{I_i} = -22.63}$

g)



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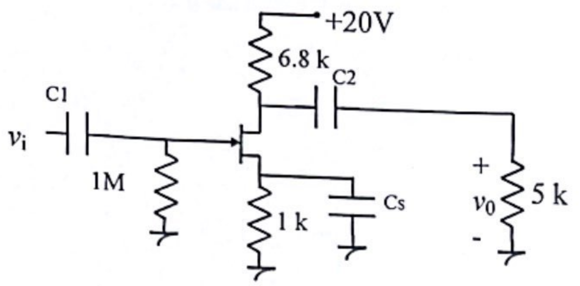
S.4 (25 puan)



I_{dss}=8mA ve V_p = −4 ise;

a) g_m=?

b) Z_i=?, Z_o=?

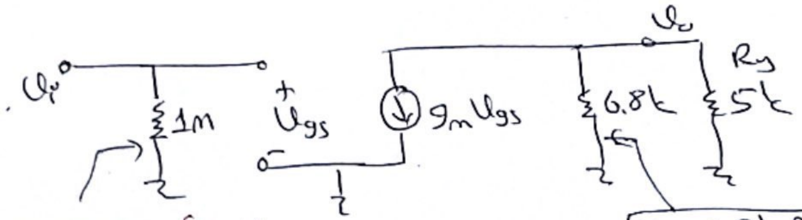
c) A_v=?, A_i=?

a) $1M.I_G + V_{GS} + I_D.1k = 0 \Rightarrow I_D = -\frac{V_{GS}}{1k} = 8m\left(1-\frac{V_{GS}}{-4}\right)^2$

$V_{GS_1} = -8V$ $\boxed{V_{GS_2} = -2V}$

$g_m = \left|\frac{2.I_{DSS}}{V_p}\right|\left(1-\frac{V_{GS}}{V_p}\right) \Rightarrow \boxed{g_m = 2mS}$ (~~8p~~) (7p)

b)



$\boxed{Z_i = 1M\Omega}$ (3p) $\boxed{Z_o = 6,8k\Omega}$ (3p)

c) $A_v = -\frac{g_m \cancel{v_{gs}} (6,8k // 5k)}{\cancel{v_{gs}}}$

$A_v = -2m . 2,88k \Rightarrow \boxed{A_v = -5,76}$ (8p)

$A_i = -A_v.\frac{Z_i}{R_y} = \frac{5,76.1000\cancel{k}}{5\cancel{k}} \Rightarrow \boxed{A_i = 1152}$ (4p)